Subnet Allocation for Smart Campus – 192.168.100.0/24

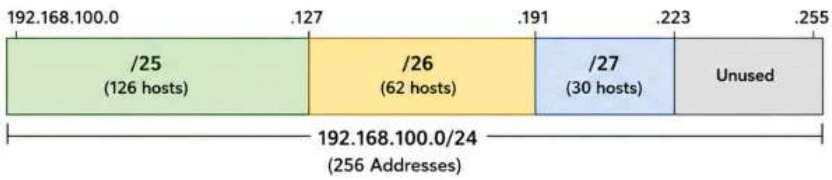
Given Network Block: **192.168.100.0/24** (255.255.255.0)

Department	Subnet / Mask	Network Address	Broadcast Address	Valid Host IP Address Range	Usable Host Addresses
School of Computing	/25 (255.255.255.128)	192.168.100.0	192.168.100.127	192.168.100.1 – 192.168.100.126	126
School of Electronics	/26 (255.255.255.192)	192.168.100.128	192.168.100.191	192.168.100.129 – 192.168.100.190	62
School of Mechanical Engineering	/27 (255.255.255.224)	192.168.100.192	192.168.100.223	192.168.100.193 – 192.168.100.222	30

1 Summary of Usable Hosts

Department	Usable Host Addresses
School of Computing (/25)	126
School of Electronics (/26)	62
School of Mechanical Engineering (/27)	30
Total Usable Hosts Allocated	**218**

2 IP Address Utilization



3 Remaining Unused IP Address Range

Unused IP Address Range:
192.168.100.224 – 192.168.100.255
Total Unused Addresses: 32

Details:

- Network Address: 192.168.100.224 (/27 network)
- Broadcast Address: 192.168.100.255
- Usable Addresses: 192.168.100.225 – 192.168.100.254 (30 addresses)
- Total Addresses in this block: 32

Overall Allocation

- Total Addresses in /24 block: 256
- Used Addresses: 224
- Unused Addresses: 32

Subnet Allocation for Company – 10.10.0.0/16 (255.255.0.0)

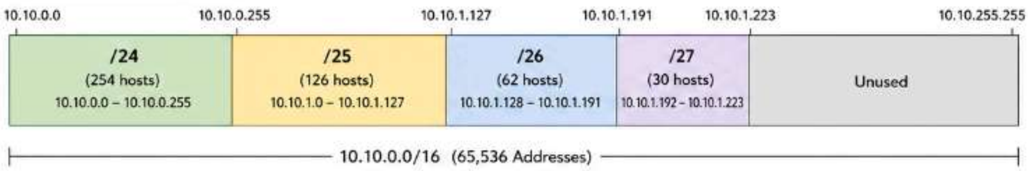
Given Network Block: 10.10.0.0/16 (255.255.0.0)
Total Addresses: 65,536 | Usable Addresses: 65,534 (10.10.0.1 – 10.10.255.254)

Department	Subnet / Mask	Network Address	Broadcast Address	Valid Host IP Address Range	Usable Host Addresses
Software Development	/24 (255.255.255.0)	10.10.0.0	10.10.0.255	10.10.0.1 – 10.10.0.254	254
Quality Assurance	/25 (255.255.255.128)	10.10.1.0	10.10.1.127	10.10.1.1 – 10.10.1.126	126
Human Resources	/26 (255.255.255.192)	10.10.1.128	10.10.1.191	10.10.1.129 – 10.10.1.190	62
Executive Management	/27 (255.255.255.224)	10.10.1.192	10.10.1.223	10.10.1.193 – 10.10.1.222	30

1. Summary – Usable Hosts

Department	Usable Host Addresses
Software Development (/24)	254
Quality Assurance (/25)	126
Human Resources (/26)	62
Executive Management (/27)	30
Total Usable Hosts Allocated	**472**

2. IP Address Utilization



Usable Range: 10.10.0.1 – 10.10.255.254

3. Remaining Unused IP Address Range

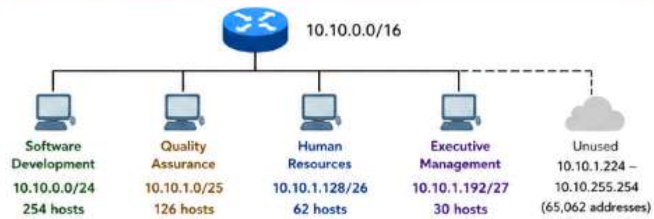
Unused IP Address Range:
10.10.1.224 – 10.10.255.254

- Network Address: 10.10.1.224 (/27 network)
- Broadcast Address: 10.10.255.255
- Usable Addresses: 10.10.1.225 – 10.10.255.254
- Total Unused Addresses: 65,534 – 472 = 65,062

4. Overall Allocation Summary

- Total Addresses in /16 block : 65,536
- Usable Addresses in /16 block : 65,534
- Used (Allocated) Addresses : 472
- Unused Addresses Remaining : 65,062
- Utilization Percentage : 0.72 %
- Unused Percentage : 99.28 %

5. Allocation Visually (Subnet Map)



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IPv6 Address Analysis for Smart Hospital

Given IPv6 Network (Full Notation):

2001:0db8:abcd:0000:0000:0000:0000:0000/64

- **Type:** IPv6 Network
- **Prefix Length:** /64

1 Compressed Notation

Convert the given IPv6 address into its compressed notation.

2001:db8:abcd::/64

Compression Rules Used:

- Leading zeros in each hextet are removed.
- The longest consecutive sequence of 0 hextets is replaced with a double colon (::) (used only once).

2 Full Hexadecimal Representation

Expand the compressed address back into its full hexadecimal representation.

2001:0db8:abcd:0000:0000:0000:0000:0000/64

Hextet Index	1	2	3	4	5	6	7	8
Hextet Value	2001	0db8	abcd	0000	0000	0000	0000	0000

- Each hextet has 4 hexadecimal digits, separated by colons.

3 Network Prefix (/64)

Identify the network prefix using the /64 prefix length.

Network Prefix (First 64 bits)

2001:0db8:abcd:0000::/64

(First 4 hextets)

/64 means first 64 bits (4 hextets) is the network prefix

Network Prefix (64 bits)	Interface ID (64 bits)
2001:0db8:abcd:0000	0000:0000:0000:0000

4 Total Number of Host Addresses Supported within the Subnet

In IPv6, with a /64 prefix:

Number of Host Bits = 128 − 64 = 64 bits

Total Number of Host Addresses = 2^{64}

= 18,446,744,073,709,551,616

★ These are the usable addresses (excluding the network prefix).

5 Total Number of Possible Addresses in the Network

Total address space in IPv6 with 128 bits:

Total Possible Addresses = 2^{128}

= 340,282,366,920,938,463,463,374,607,431,768,211,456

Addresses in this /64 Subnet:

2^{64}

= 18,446,744,073,709,551,616

Note:

The /64 subnet is a part of the entire IPv6 address space.

Routing Table Analysis – Packet Destination 192.168.2.45

Given Routing Table Entries:

192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

Destination IP Address of Packet: 192.168.2.45

How /24 Works

- /24 means the first 24 bits (3 octets) are the network portion.
- Subnet mask: 255.255.255.0
- So, any IP from 192.168.2.1 to 192.168.2.254 belongs to **192.168.2.0/24**

1 Destination Subnet

The destination IP **192.168.2.45** belongs to the subnet:

192.168.2.0/24

Range of this subnet: 192.168.2.1 – 192.168.2.254

Network Address: 192.168.2.0

Broadcast Address: 192.168.2.255

2 Matching Routing Table Entry

Routing Table	Network	Subnet Mask	Network Address	Match with 192.168.2.45	Result
Entry 1	192.168.1.0	255.255.255.0	192.168.1.0	No	No Match
Entry 2	**192.168.2.0**	**255.255.255.0**	**192.168.2.0**	**Yes**	**Match ✓**
Entry 3	192.168.3.0	255.255.255.0	192.168.3.0	No	No Match

The destination IP matches with the routing table entry:

192.168.2.0/24

3 Next-Hop Network / Interface

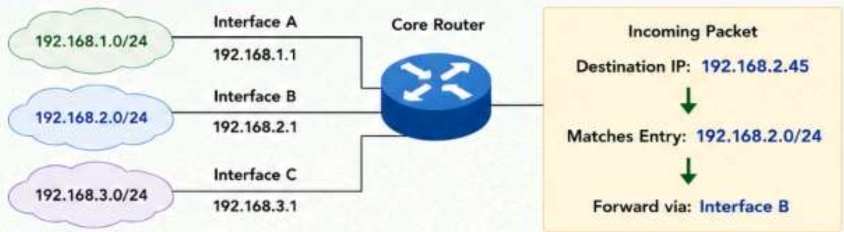
The packet will be forwarded using the next-hop interface associated with the matching entry (192.168.2.0/24).

Next-Hop Interface Used:

Interface B

(Connected to 192.168.2.0/24 network)

Routing Decision Process



Summary

1. Destination Subnet	2. Matching Routing Table Entry	3. Next-Hop Network / Interface
192.168.2.0/24	192.168.2.0/24	Interface B (192.168.2.0/24)

Default Route – Used When No Match Found

Definition: The default route is used by the router when the destination IP address does not match any specific routing table entry.

Default Route Entry in Routing Table:
0.0.0.0/0

1. Routing Table (Including Default Route)

Destination Network	Subnet Mask / Prefix	Next-Hop / Interface	Type
192.168.1.0	255.255.255.0 (/24)	Interface A	Specific
192.168.2.0	255.255.255.0 (/24)	Interface B	Specific
192.168.3.0	255.255.255.0 (/24)	Interface C	Specific
0.0.0.0	0.0.0.0 (/0)	Interface D (ISP)	Default

0.0.0.0/0 matches every possible IP address.
It is the **"catch-all" route.**

2. Example Scenario

Routing Table contains only the entries shown on the left.

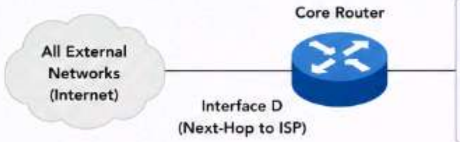
Incoming Packet Destination IP: 8.8.8.8

Does it match any specific route?

- 8.8.8.8 is not in 192.168.1.0/24
- 8.8.8.8 is not in 192.168.2.0/24
- 8.8.8.8 is not in 192.168.3.0/24

No match found in specific routes.
Therefore, the router uses the **default route (0.0.0.0/0)**.

3. How the Default Route Works



Internal Networks
192.168.1.0/24 (Interface A)
192.168.2.0/24 (Interface B)
192.168.3.0/24 (Interface C)

If the destination does not match any internal network,
the packet is forwarded to the ISP via **Interface D (default route)**.

4. Summary

Default Route	0.0.0.0/0
Purpose	Used when no specific route matches the destination IP.
Matches	All IP addresses
Next-Hop / Interface	Interface D (ISP)
Example Packet Destination	8.8.8.8
Action Taken	Forwarded via Default Route **(0.0.0.0/0)**

Key Point

★ The default route ensures that packets for unknown or external destinations can still be forwarded, preventing traffic loss when no specific route exists.

Default Route Entry: 0.0.0.0/0 → Interface D (ISP)

5. A Company Establishes Two Office Departments Connected through a Layer-3 Device (Router)

A company has two office departments, HR and IT, connected through a Layer-3 device (Router). Each department has its own subnet.
Design the IP addressing scheme and show how the two departments can communicate with each other.

1. Network Requirements

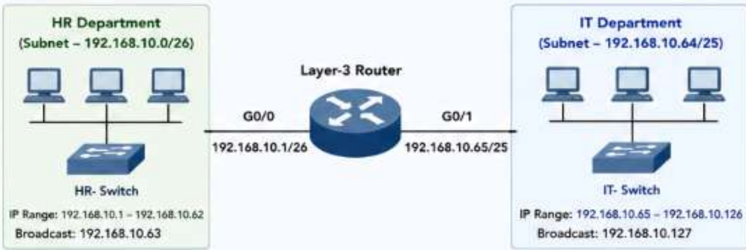
- HR Department : 50 Hosts
- IT Department : 100 Hosts
- Inter-department communication through Router (Layer-3)

2. IP Addressing Plan

Network	Subnet / Mask	Network Address	Usable Host Range	Broadcast	Usable Hosts
HR Department	/26 (255.255.255.192)	192.168.10.0	192.168.10.1 – 192.168.10.62	192.168.10.63	62
IT Department	/25 (255.255.255.128)	192.168.10.64	192.168.10.65 – 192.168.10.126	192.168.10.127	126
Router Links (G0/0 & G0/1)	N/A	N/A	N/A	N/A	N/A

Note: 192.168.10.0/24 network is used and divided using VLSM.

3. Network Topology



The router acts as the default gateway for both departments.

4. Router Interface Configuration

Interface	IP Address	Subnet Mask	Connected To
GigabitEthernet0/0 (To HR Department)	192.168.10.1	255.255.255.192 (/26)	HR Switch
GigabitEthernet0/1 (To IT Department)	192.168.10.65	255.255.255.128 (/25)	IT Switch

Example Configuration (Cisco IOS)

```
interface GigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.192
 no shutdown
!
interface GigabitEthernet0/1
 ip address 192.168.10.65 255.255.255.128
 no shutdown
```

5. Communication Between Departments

- HR PC sends packet to IT PC.
- Since destination (192.168.10.80) is not in the HR subnet (192.168.10.0/26), the packet is sent to the default gateway (192.168.10.1).
- Router routes the packet out of G0/1 interface to the IT subnet (192.168.10.64/25).
- IT PC receives the packet and reply follows the same path in reverse.

6. Routing Table on Router

Destination Network	Subnet Mask	Next Hop	Exit Interface
192.168.10.0	255.255.255.192	Directly Connected	G0/0
192.168.10.64	255.255.255.128	Directly Connected	G0/1
0.0.0.0	0.0.0.0	N/A	N/A

Both networks are directly connected, so no static routes are required.

7. Summary

- Two departments are assigned different subnets using VLSM.
- Router interfaces act as default gateways for their respective subnets.
- Layer-3 routing enables communication between the two departments.
- This design is scalable and secure.

Department	Subnet / Mask	Network Address	Usable Host Range	Broadcast Address	Default Gateway (Router Interface)
HR Department	/26 (255.255.255.192)	192.168.10.0	192.168.10.1 – 192.168.10.62	192.168.10.63	192.168.10.1 (G0/0)
IT Department	/25 (255.255.255.128)	192.168.10.64	192.168.10.65 – 192.168.10.126	192.168.10.127	192.168.10.65 (G0/1)